

CSA 1718 – Artificial Intelligence

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Experiment: Prolog Database for Name and Date of Birth

Aim

To write and execute a Prolog program to create a simple database containing a person's **name and date of birth (DOB)** and retrieve the stored information using queries.

Objective

- To understand facts in Prolog.
- To create a simple database using Prolog.
- To retrieve information using Prolog queries.

Theory

A Prolog database can be created using **facts**. Each person can be represented using a predicate such as:

person(Name, DOB).

Here, Name represents the person's name and DOB represents the date of birth.

Prolog Program

% Database containing Name and Date of Birth

person(ravi, date(15, 5, 2005)).

person(anu, date(20, 8, 2006)).

person(kiran, date(10, 12, 2004)).

person(priya, date(25, 3, 2007)).

person(arun, date(5, 11, 2005)).

Sample Queries

1. Find the DOB of Ravi

?- person(ravi, DOB).

Output

DOB = date(15, 5, 2005).

2. Find the DOB of Priya

?- person(priya, DOB).

Output

DOB = date(25, 3, 2007).

3. Find all persons in the database

?- person(Name, DOB).

Output

Name = ravi,

DOB = date(15, 5, 2005) ;

Name = anu,

DOB = date(20, 8, 2006) ;

Name = kiran,

DOB = date(10, 12, 2004) ;

Name = priya,

DOB = date(25, 3, 2007) ;

Name = arun,

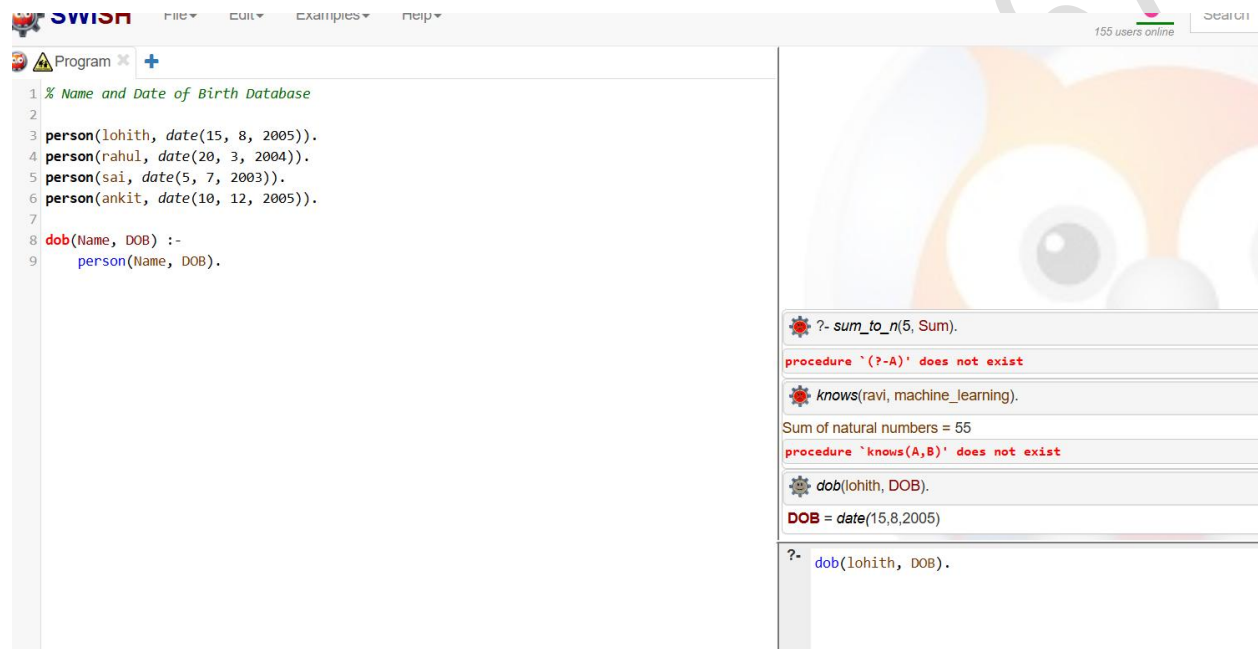
DOB = date(5, 11, 2005).

Algorithm

1. Start the program.
2. Define the person(Name, DOB) predicate.

3. Store names and corresponding dates of birth as facts.
4. Enter a query to retrieve a person's DOB.
5. Prolog searches the database for a matching fact.
6. Display the name and DOB.
7. Stop.

Out Put



The screenshot shows the SWISH Prolog IDE interface. On the left, a code editor displays a Prolog program with the following content:

```
1 % Name and Date of Birth Database
2
3 person(lohith, date(15, 8, 2005)).
4 person(rahul, date(20, 3, 2004)).
5 person(sai, date(5, 7, 2003)).
6 person(ankit, date(10, 12, 2005)).
7
8 dob(Name, DOB) :-
9     person(Name, DOB).
```

On the right, the execution output is shown, displaying several failed queries and one successful query:

```
?- sum_to_n(5, Sum).
procedure `(?-A)' does not exist

?- knows(ravi, machine_learning).
Sum of natural numbers = 55
procedure `knows(A,B)' does not exist

?- dob(lohith, DOB).
DOB = date(15,8,2005)

?- dob(lohith, DOB).
```

Result

The Prolog program was successfully created and executed to store and retrieve **name and date of birth information** using a simple Prolog database.